

Existence Proofs

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Outline

When One Example is Enough

Splitting an Octagon

Making Fun in Real Life

Know Your Rights

Nobody Can Win All The Times

Know What Are You Looking For



Source: [https://en.wikipedia.org/wiki/Historia_animalium_\(Gessner\)](https://en.wikipedia.org/wiki/Historia_animalium_(Gessner))

Proofs For Existential Statements

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- one example is enough

Cutting Figures

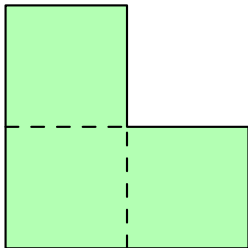
Cutting Figures

congruent pieces: of the same shape and size

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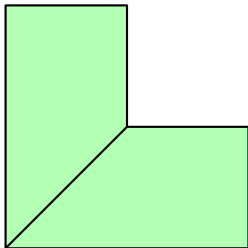
Prove that this figure can be cut into 2 congruent pieces



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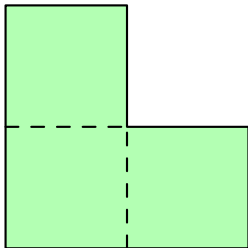
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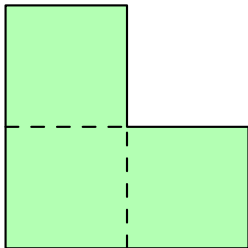
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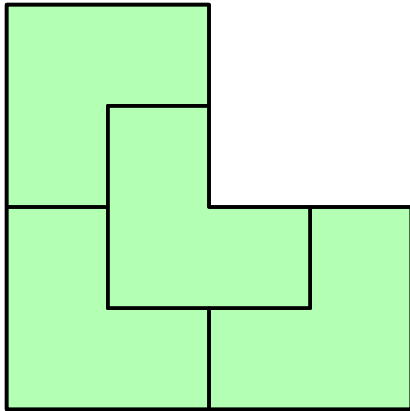
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what about 4 pieces?

Spoiler

Spoiler



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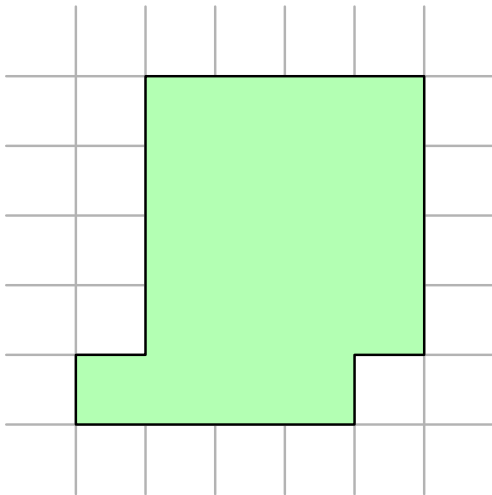
Making Fun in Real Life

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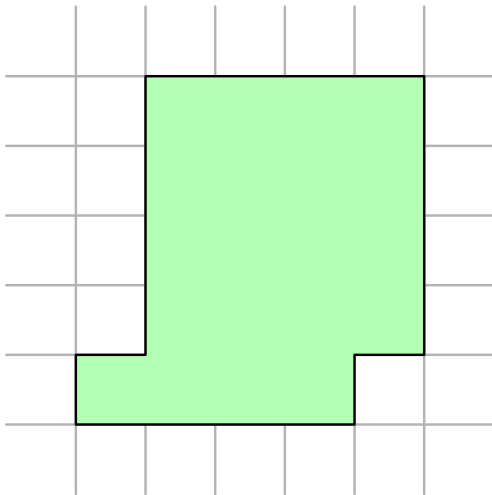
Nobody Can Win All The Times

The Octagon

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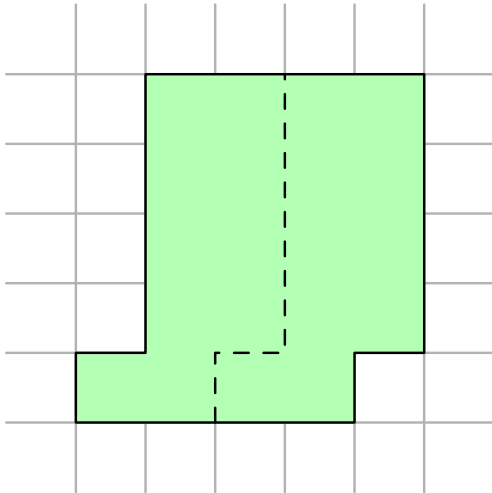
The Octagon



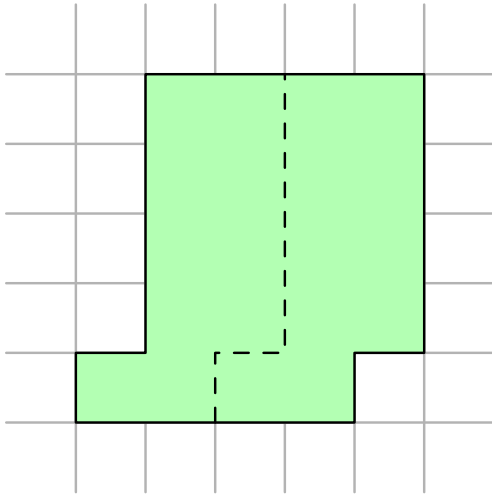
split into two congruent pieces

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what about three congruent pieces?

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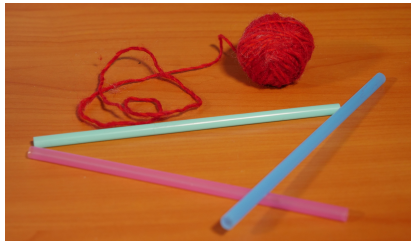
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Tensegrities

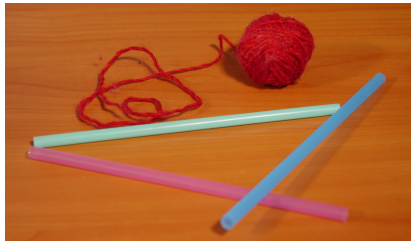
Tensegrities

- drinking straws and thread



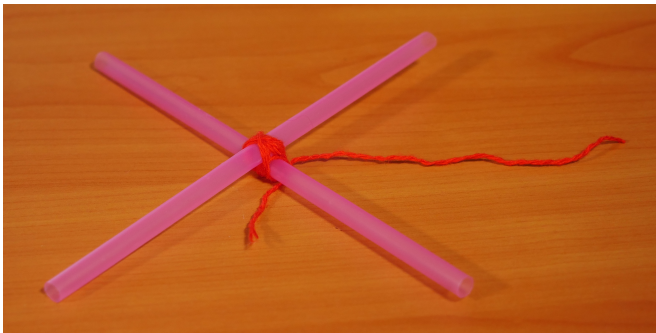
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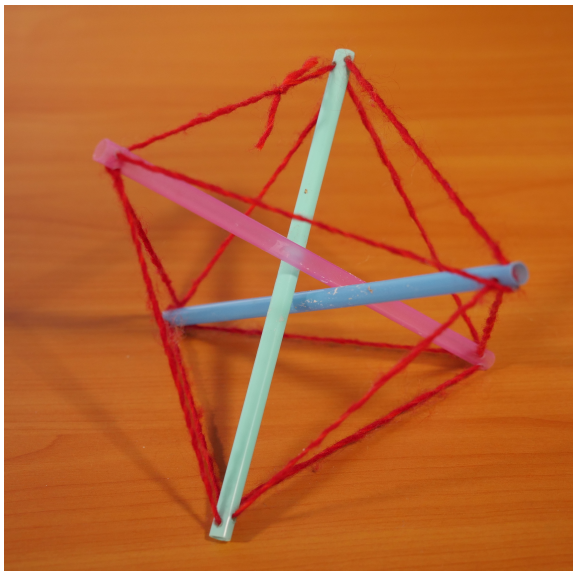


- there exists a “tensegrity”: a solid construction; straws do not touch each other; connected by threads

Not Allowed



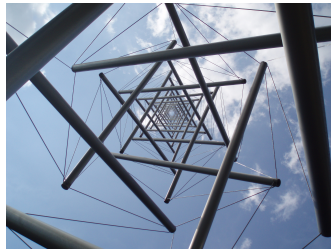
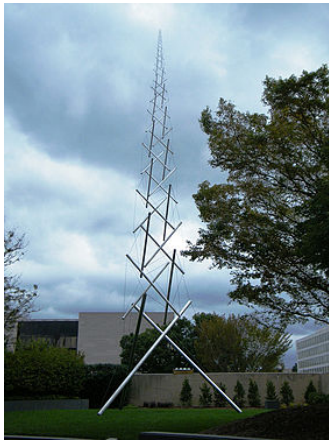
Tensegrity Finished



A Tensegrity: Animation

[Source: https://commons.wikimedia.org/wiki/File:Tensegrity_simple_3.gif]

Tensegrities in the Real Life



[Source:https://en.wikipedia.org/wiki/Needle_Tower]

made by Kenneth Snelson, a student of Buckminster Fuller (who invented the word and made many of them)

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- you say: $7125 = 57 \cdot 125$

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- divisible by 7: 14, 21, 28, 35, 42, 49, 56, 63
- but what if we asked for a number that becomes 57 times smaller?
- you say: $7125 = 57 \cdot 125$
- no need to explain how you found it

No Rights for Teachers

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- a divisible by 7: only $a = 7$ works
- $10^k = 8 \times X$; 10 and 100 not multiples of 8
- 1000 works, $X = 125$
- also $71250 = 57 \times 1250$, etc.

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- $+1 + 2 - 3 = 0$

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- total weight 21: not a multiple of 2
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- what about weights 2, 4, 6, 8, 10, 12?
- hint: just changing the units

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- NP-complete $\approx$ infeasible

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- claim: *an object with some property exists*
- proof: *an example*
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- no need to disclose the sources
- beware: claim may be false!